



FIELD DEMONSTRATION AND EVALUATION OF TERRESTRIAL NAVIGATION TECHNOLOGIES FOR ARTEMIS SURFACE NAVIGATION APPLICATIONS

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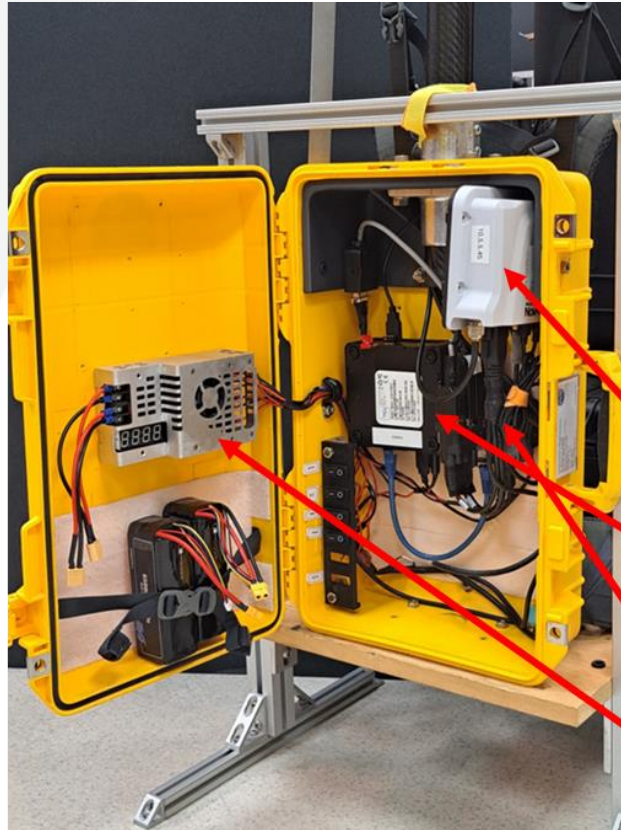
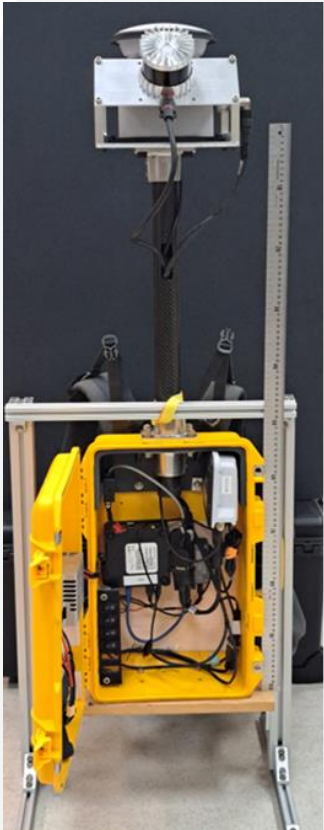
Outline

- Overview of Potential Surface Navigation Approaches
 - LIDAR-based Navigation with KNaCK
 - Beacon-based Localization via Lunar Harpoon
 - Visual Inertial Navigation using VOXL-CAM
 - Photogrammetry using COTS Imagers
- Key Potential Use Cases
- Cross-comparison of Solutions
- Focused Application of Technologies
- Conclusions

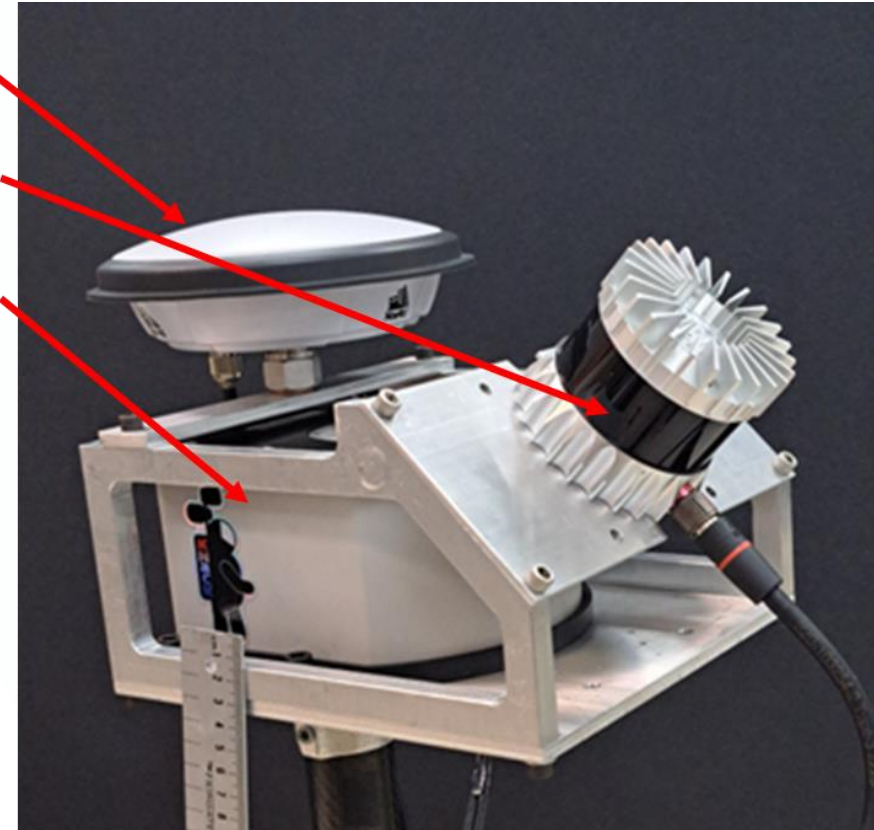
Kinematic Navigation and Cartography Knapsack (KNaCK)



- Person-mounted mobile LiDAR scanning system
- Ouster OS-1 LiDAR Sensor, HG1700 IMU, GNSS ground-truth
- Implementation of SLAM and TRN algorithms



GNSS
antenna
OS-1 LiDAR
Sensor
HG1700
IMU
PwrPak 7 INS
computer
Intel NUC
computer
Ouster
Interface Box
Power Supply



The KNaCK SLAM w/ TRN Algorithm Performance

SLAM: Simultaneous Localization and Mapping
TRN: Terrain Relative Navigation

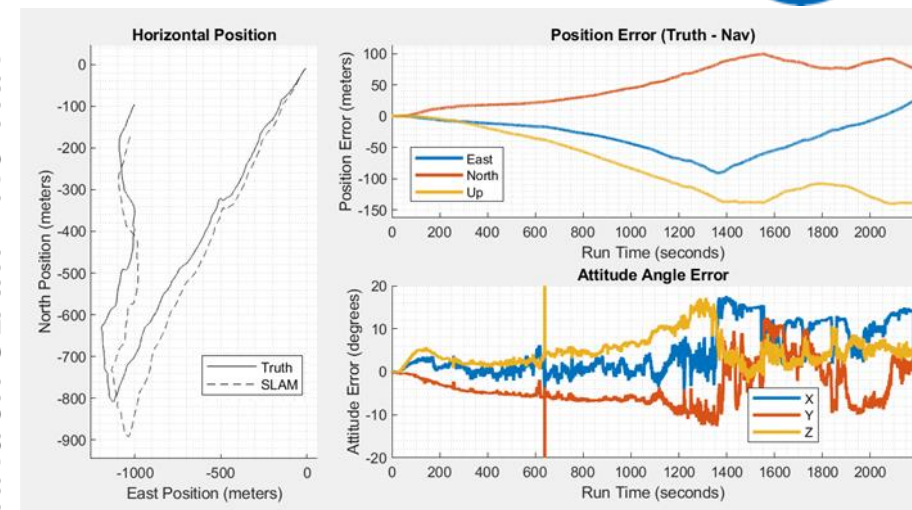
KNaCK SLAM currently provides:

Real-time processing for Navigation Track from FMCW-LiDAR 6-DoF

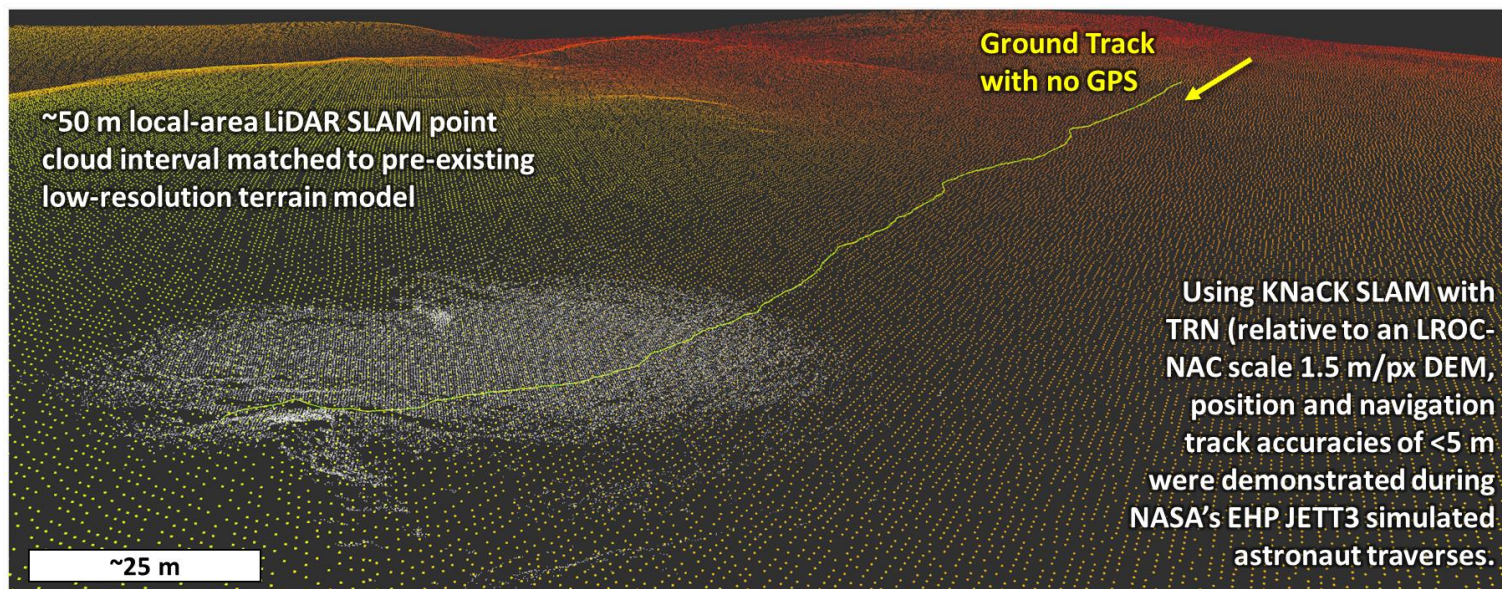
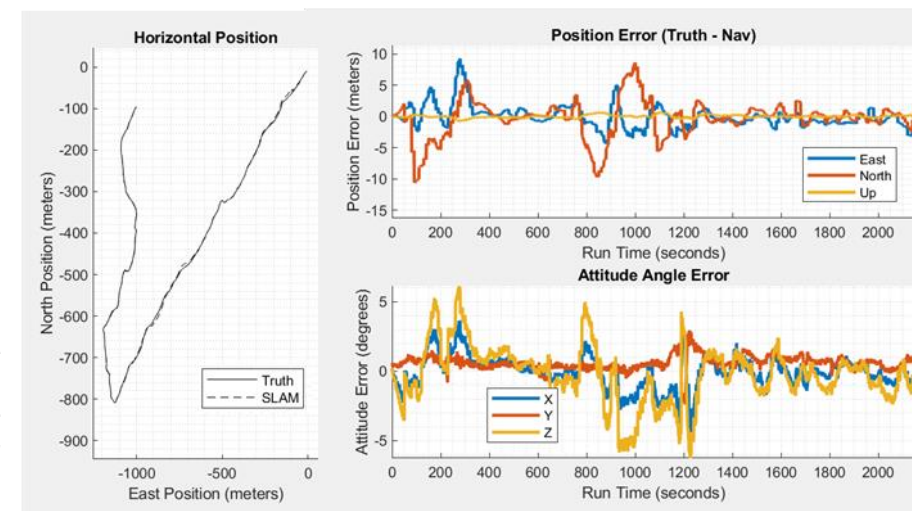
- Accurate GPS-denied mapping
- Verification of Real-time Nav Data versus GPS
- SLAM Loop Closure
- Tested to better than 5 m location accuracy with TRN (compared to GPS-ground-truth)



KNaCK-SLAM – No TRN



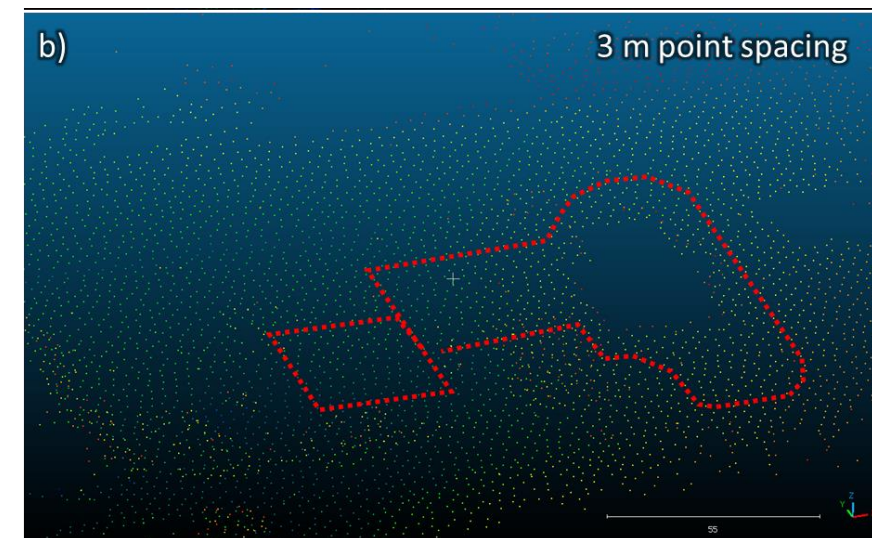
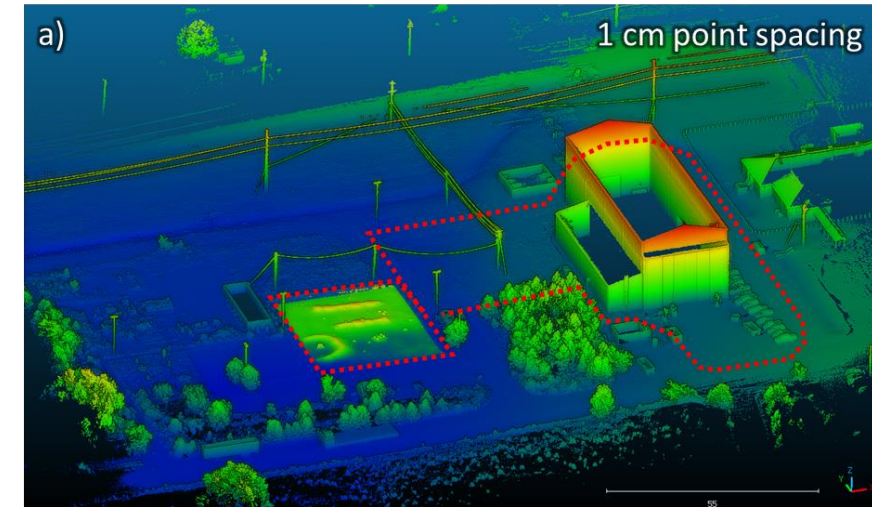
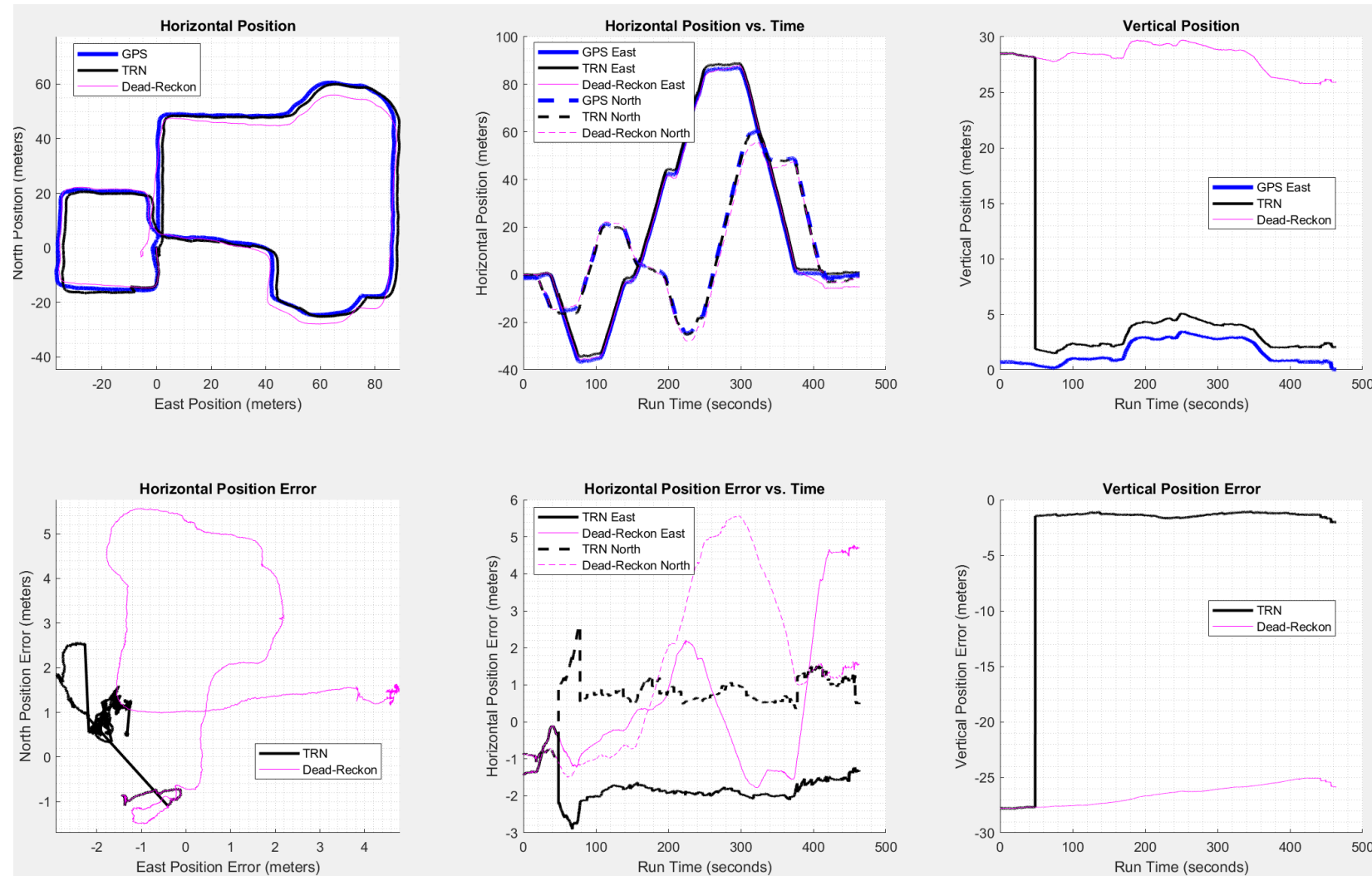
KNaCK-SLAM – with TRN



TRN Basemaps and Ground-testing



- Aim to replicate Lunar Reconnaissance Orbiter Camera (LROC) digital elevation model (DEM) resolution ($\sim 1\text{-}3\text{m/pixel}$)



MSFC's Lunar Regolith Terrain (LRT)



Figure 1. Oblique view of MSFC's Lunar Regolith Terrain (LRT) field

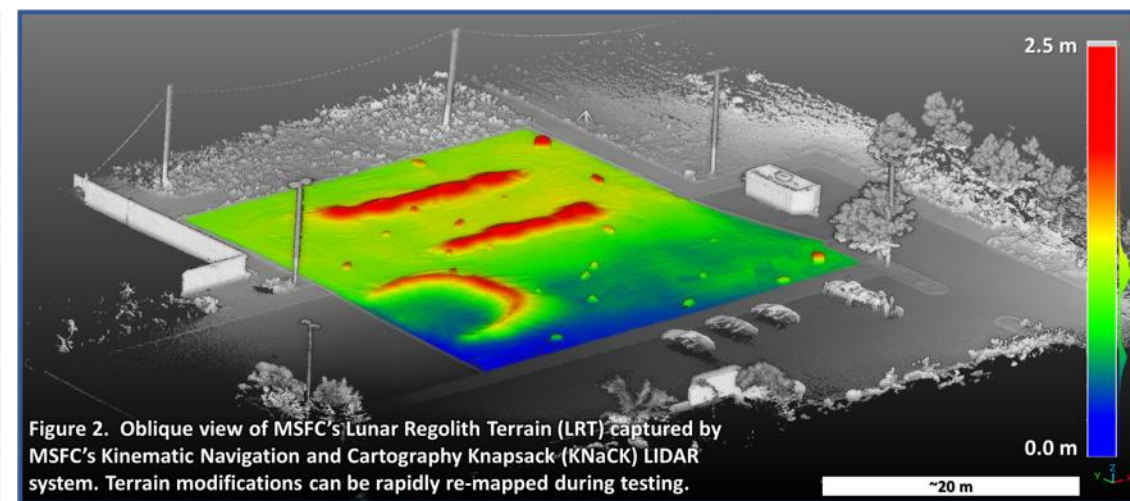


Figure 2. Oblique view of MSFC's Lunar Regolith Terrain (LRT) captured by MSFC's Kinematic Navigation and Cartography Knapsack (KNaCK) LIDAR system. Terrain modifications can be rapidly re-mapped during testing.



Figure 3. Removable Full-size Pallet Lander Mockup for rover deployment

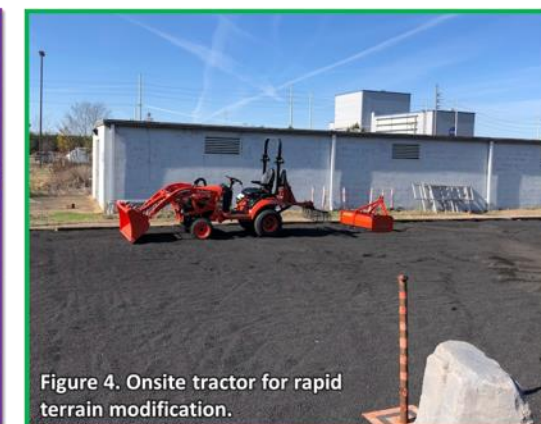


Figure 4. Onsite tractor for rapid terrain modification.

Lunar Regolith Terrain (LRT) Description:

The Lunar Regolith Terrain field is an outdoor planetary analog environment facility located on base at MSFC:

- Applies a lunar regolith simulant of JSC-1A feedstock material (volcanic cinder sand sourced from Meriam Crater, Flagstaff, AZ) with representative geotechnical, geochemical, and optical properties of lunar mare basalt.
- Contains 500 tons of lunar regolith simulant confined within a 125 ft x 125 ft (38 m x 38 m) area with a depth range between ~5 in - ~4 ft (~13 cm - 1.2m) that can be modified to suit user needs.

Current Features of the LRT:

The LRT was designed to allow rapid modification of the terrain's topography obstacles in the field:

- The terrain can be reshaped to suite specific testing requirements that may require flat-expanses, steep hills, or heavily cratered and rocky landscapes.
- Large rocky obstacles in Fig. 1 are artificial landscape boulders (faux-rocks) that can be easily placed by users or removed entirely.
- Rapid modification capabilities for burial of additional user-specific materials to enable in-situ resource utilization detection (e.g., burial of hydrogen sources for neutron detection or other materials).

Upcoming Features of the LRT:

Development of the LRT's capabilities continues:

- Direct radio frequency communication with the Huntsville Operations Support Center (HOSC).

Currently Available to the Community:

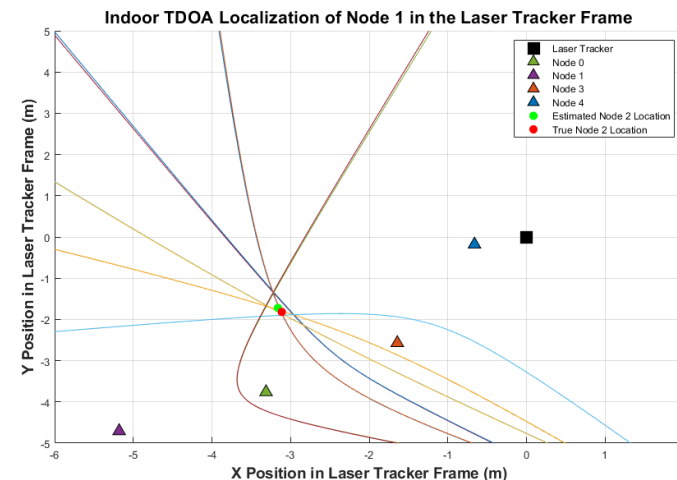
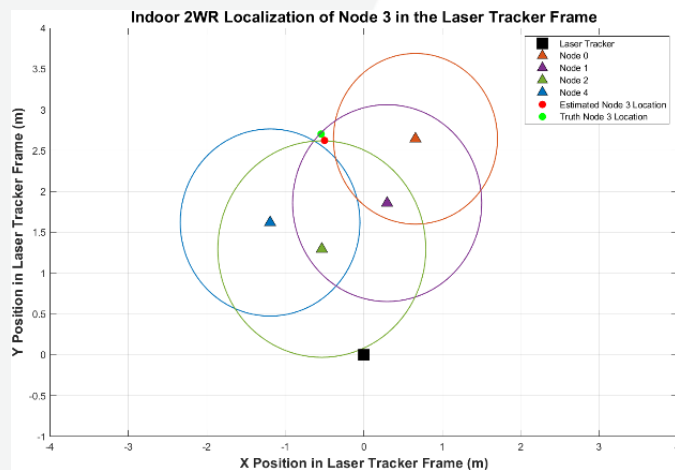
- The LRTF provides an accessible planetary analog surface environment for surface mobility testing, autonomous roving operations, developing advanced navigation techniques and operations development.

Meets the needs of visiting research groups:

- Equipped with on-site office space: an air-conditioned and heated trailer with 120/240V power and lighting.
- Wi-Fi and Cellular signal coverage.
- On-site workspaces and secure equipment storage is available in adjacent buildings.
- Accessibility for on-site parking and delivery of instruments, payloads, and additional equipment.

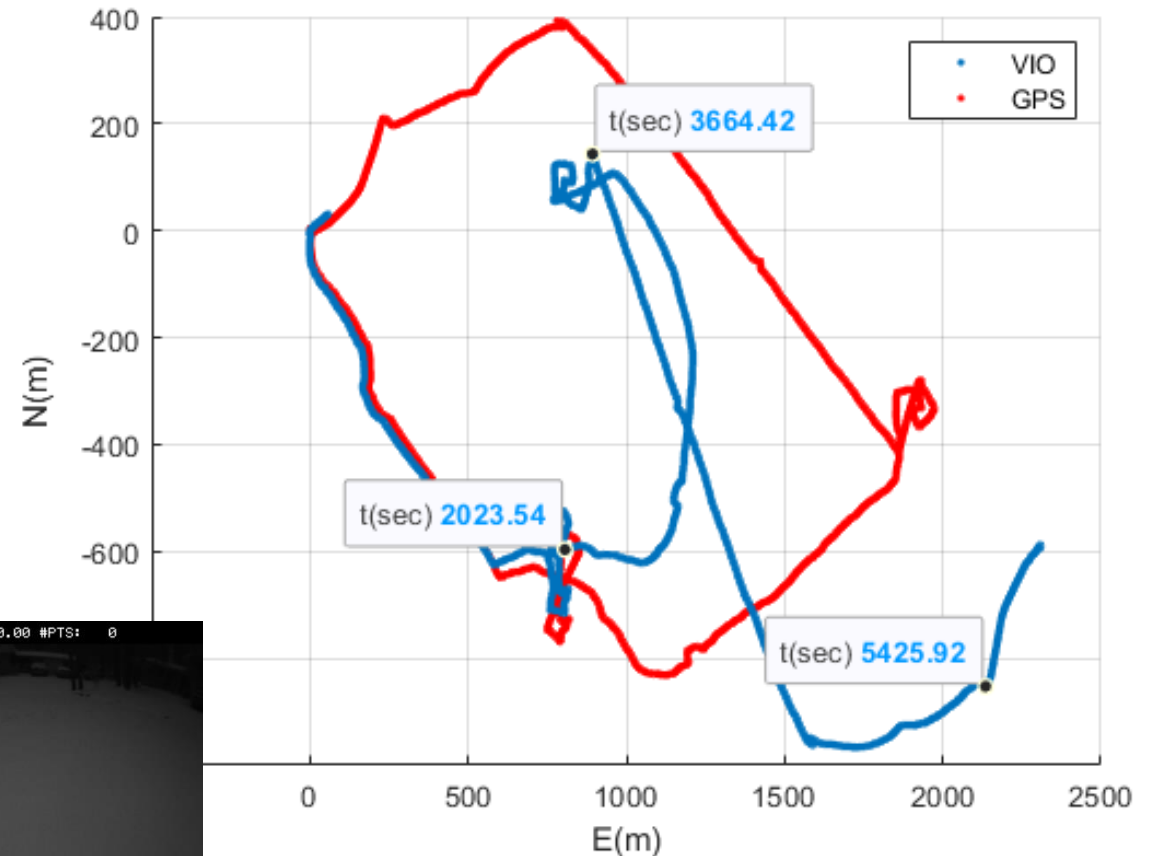
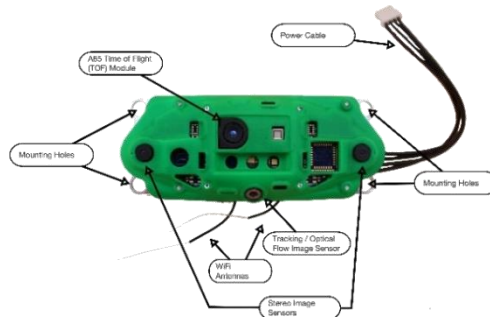
Lunar Harpoon – Utilization of UWB Beacons

- Deployable psuedolite arrays
 - 2WR, TDOA, PDOA
 - ~10 cm accuracies
- 2.4GHz comm, UWB nav, GPS-RTK
- Range limited to 50m
 - Testing upgraded sensors to 500 m



Visual Inertial Odometry via VOXL-CAM Platform

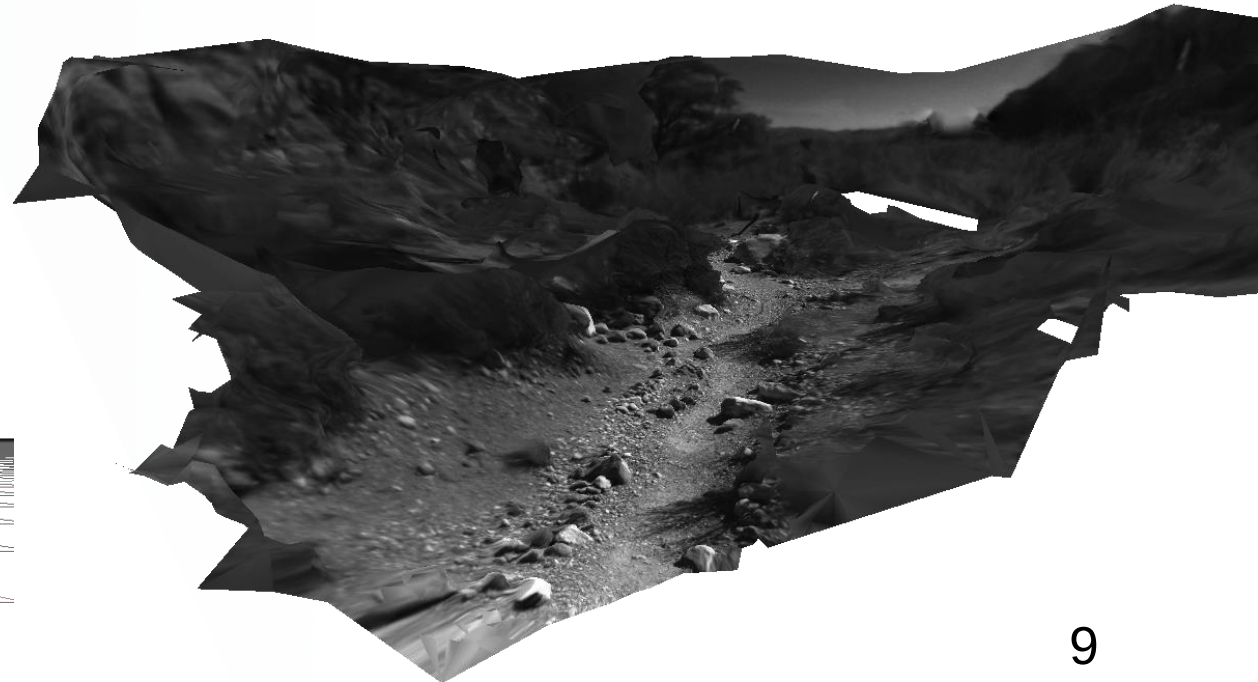
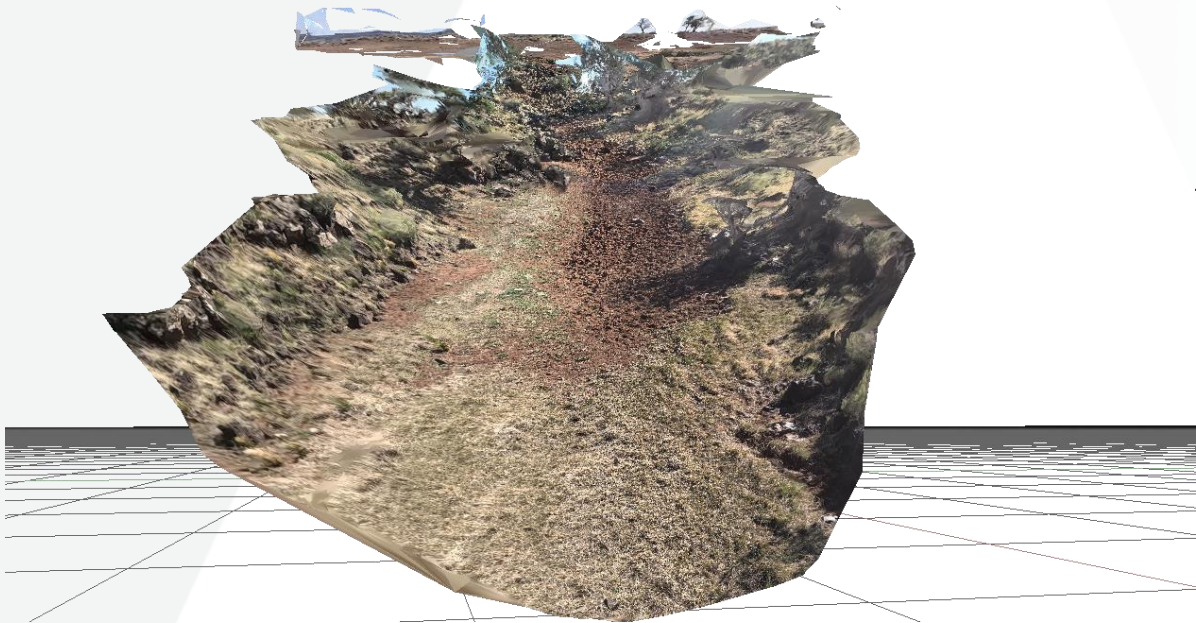
- Originally developed for drone and GPS-denied scenarios by ModalAI
- Computational platform has heritage to Ingenuity
- Integrates onboard sensors to provide a dead reckoning solution
- Attitude and distances augmented by visual odometry



Example Results form JETT-7

Photogrammetry Using OTS Imagers

- Originally developed for model generation can also be used for terrain reconstruction with applications in archaeology and surveying (among others)
- GoPro Hero 11 results (L) for data generation at 4K and 1080p
 - Synchronized imagery, accelerometer, gyroscope, and GPS
- VOXL-CAM wide-angle Tracking Camera also used (R)
- OTS processor (3DFZephyr) used to process 10Hz images to generate overall scene
- Heavy computational process, tries to tie every image to a global scene



Key Use Cases



Use Case	Description	Coarse Timeline
Near-field Exploration (100s of meters)	Ability for the crew to understand their location (absolute or relative) and being able to return to a host vehicle or asset	Early (Artemis III+)
Absolute Localization	Ability to correlate a sensor or other scientific observation (including imagery) to an absolute location on the lunar surface, with shift from a post-processing to real-time capability.	Early to Mid (Artemis IV+)
Far-field Navigation (km's)	Ability for the crew to conduct an expanded traverse around a fixed surface element (i.e., HLS) to both reach scientific areas of interest safely return to the host vehicle.	Mid (Artemis V)
Multi-asset Autonomous Operations	Ability for multiple assets to understand precision relative states to support data cross correlation and coordinated scenarios such as multiple rovers.	Late (Artemis VI+)
Large Permanently Shadowed Region	Ability for the crew or a vehicle to traverse independently into a large, shadowed area where line of sight to other surface elements and the sun is not available.	Late (Artemis VI+)



Comparison of Performance

Approach	Near-field Exploration	Absolute Localization	Far-field Navigation	Multi-asset Autonomous Operations	Large Permanently Shadowed Region
LIDAR-based Navigation	High	High	High	Medium	High
Local Radiometric Network	Medium	High	Low	High	Medium
Visual Inertial Odometry	Medium	Medium	Medium	Low	Medium
Monocular SLAM Techniques	Low	High	Medium	Low	Low
Photogrammetry	Low	High	Low	Low	Low

- No solution universally out-performing, all involve trade-offs between performance, use-case, and onboard processing capability



Highlighted Applications of Each

- **KNACK:** navigation into regions where have a-priori mapping products and no guarantees of lighting conditions, fully self-contained system provide ~m level performance
- **Harpoon:** high accuracy needed in a local area, i.e. construction, formation driving, autonomous robotics
- **Visual Inertial Odometry:** can help to reduce growth of inertial errors during periods of outages from other vehicles, ability to embed within small platforms
- **Photogrammetry:** reconstruction and development of detailed surface maps between traverses, and post-processing



Conclusion and Summary

- Multiple technologies offer a path towards improved surface navigation for crew
- Capabilities for all demonstrated through field testing
- Forward work
 - Continue efforts to validate hardware for space environment (thermal and radiation primary concerns)
 - For VIO: develop more mature front-end to maintain solution during odometry drop-outs, and incorporate heading measurements via sun-sensor
 - For Harpoons: integrate longer-range sensors to expand operational areas
 - For photogrammetry: mature process for building up and processing batches of imagery over local areas to stitch together into larger model
 - Continue to assess other technologies: radio direction finding standalone solutions, positioning using WiFi signals (RTT and steering matrix), and UWB AoA solutions

Thank you very much! Any questions?

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The authors would like to acknowledge and express thanks to all of the field testing teams, specifically MSFC's LRT, the JETT team, and the Mars Institute. Thank you to all of the engineering teams who have helped to implement, test, and assess these technologies!

